National Bamboo Shoots Value Chain Development Programme

Proposal for Government Partnership and Financial Support

Submitted to: The Honourable Floyd Green, MP, Minister of Agriculture, Fisheries and Mining, Jamaica

Submitted by: Bamboo Industry Association of Jamaica (BIAJ)

Proposed duration: 24 months | Proposed funding: J$49,588,000 | Date: October 2026

STATUS: Board-review draft. All financial figures are indicative planning estimates, not supplier quotations or confirmed government commitments.

1. Executive summary

Following a meeting between representatives of the BIAJ Board of Directors and the Honourable Minister Floyd Green, the Association respectfully submits this proposal in response to the Minister's invitation to outline assistance needed to develop the edible bamboo shoots value chain in Jamaica.

BIAJ proposes a 24-month demonstration and commercialization programme centered initially on locally available Bambusa vulgaris. The programme will establish a controlled supply of raw shoots, validate safe food-processing methods, pilot packaged products, develop markets and strengthen farmer and small-enterprise capacity. Commercial distribution will only begin after appropriate food-safety validation and applicable regulatory clearances.

The indicative total programme cost is J$49.59 million, comprising a requested Government contribution of J$39.67 million (80%) and proposed BIAJ/private/in-kind contributions of J$9.92 million (20%), subject to partner commitments.

2. Strategic rationale

Jamaica has bamboo resources and an opportunity to create additional agricultural value through edible shoots. However, the sector requires coordinated development of planting material, agronomy, harvest logistics, validated detoxification/processing methods, quality assurance, product development, market access and business systems. This proposal seeks to convert fragmented initiatives into a traceable, safe and commercially testable supply chain.

3. Programme objectives

- Validate an appropriate, safe and reproducible process for edible shoots from locally sourced Bambusa vulgaris, including control of cyanogenic compounds.
- Organize and train an initial cohort of 30 growers/harvest suppliers and 10 processing/enterprise participants.
- Establish a demonstration supply chain and complete at least three documented pilot processing runs.
- Develop and assess at least two product formats, including a shelf-stable canned option only if validated and approved.
- Generate market evidence through structured buyer engagement, pricing trials and commercial feasibility analysis.
- Create a practical scale-up roadmap, quality manual and investment-ready business case.

4. Work packages and implementation

A. Farmer supply and field development (Months 1–18)

Map suitable bamboo stands and participating farms; document shoot availability and seasonality; develop harvesting specifications, handling protocols, collection points and traceability. Provide training and starter field support to a target cohort of 30 growers/suppliers.

B. Food safety, standards and product validation (Months 1–15)

Commission a technical review of edible-shoot processing and develop a validated method for managing cyanogenic compounds. Engage the Scientific Research Council (SRC), Bureau of Standards Jamaica (BSJ), competent food-safety authorities and independent laboratories as appropriate. Test microbiological safety, cyanogenic-compound levels, packaging integrity, shelf stability and product quality. For canned samples, include can-seam analysis. No shelf-life or safety claim will be made without adequate supporting tests.

C. Processing and packaging pilot (Months 4–18)

Undertake pilot batches using an appropriately equipped, compliant processing facility. Evaluate equipment needs, yields, water and energy use, waste management, packaging sizes, labels and unit economics. BIAJ's existing discussions concerning SRC pilot processing can provide a starting point, subject to formal agreement.

D. Business development and market testing (Months 8–24)

Assess hotels, restaurants, supermarkets, institutional purchasers and potential export distributors; commission consumer and chef trials; calculate production costs and wholesale/retail price ranges; prepare branding and a scale-up investment prospectus.

E. Governance, monitoring and reporting (Months 1–24)

Create a BIAJ-led steering committee with Ministry representation and technical partners as agreed. Establish baseline measures, procurement controls, quarterly financial and performance reports, milestone-based disbursements and independent final evaluation.

5. Indicative implementation milestones

Period	Key deliverable	Decision gate
Months 1–3	Partners, baseline, site selection, technical protocols	Approve pilot plan
Months 4–6	Grower training; initial raw-material trials	Review handling and safety data
Months 7–12	Processing trials; lab testing; package prototypes	Safety and feasibility review
Months 13–18	Validated pilot runs; controlled buyer trials	Authorize eligible market trials
Months 19–24	Business case, investment plan, final report	Decision on national expansion

6. Detailed indicative budget (JMD)

Budget assumptions: 24 months; amounts are preliminary allowances in JMD; quantities indicate planning units, not awarded contracts. Contingency is 10% of direct costs.

Cost item	Qty	Unit (J$)	Total (J$)
Programme coordination and technical management	24	220,000	5,280,000
Agronomy surveys, baseline and mapping	1	2,200,000	2,200,000
Grower training workshops and field support	12	250,000	3,000,000
Demonstration plots and collection infrastructure	3	1,400,000	4,200,000
Raw-shoot procurement and transport for trials	1	2,600,000	2,600,000
Process design and technical consultancy	1	3,500,000	3,500,000
Food safety laboratory testing and validation	1	5,200,000	5,200,000

Pilot processing facility access and operating runs	1	5,800,000	5,800,000
Small-scale processing and quality-control equipment	1	6,400,000	6,400,000
Packaging, labels and prototype development	1	2,600,000	2,600,000
Market research, buyer trials and promotion	1	2,500,000	2,500,000
Monitoring, audit and independent evaluation	1	1,800,000	1,800,000
Direct programme cost			45,080,000
Contingency (10%)			4,508,000
TOTAL PROGRAMME COST			49,588,000

Proposed funding structure: Government 80% = J$39,670,400; BIAJ/other partners 20% = J$9,917,600. The counterpart share is a target and is not yet confirmed.

7. Support requested from the Ministry

- Approve the programme in principle and designate a Ministry focal point for technical coordination.
- Consider a grant or phased allocation toward the proposed Government share, subject to due diligence and available public resources.
- Facilitate engagement with relevant agencies, research institutions, extension services and food-safety regulators.
- Support identification of participating farmers, demonstration sites and opportunities for public–private investment.
- Agree a memorandum of understanding setting out roles, reporting, procurement, safeguards and performance milestones.

8. Expected results and performance indicators

- 30 growers/harvest suppliers trained, with participation and adoption records.
- At least three pilot production runs fully documented with traceable raw material and test results.
- At least two product concepts assessed for technical and commercial feasibility.
- A documented, regulator-reviewed food-safety and quality-control pathway before commercial sale.

- At least 10 prospective institutional or retail buyers consulted; demand and pricing findings reported.
- A costed scale-up strategy and final evaluation delivered by Month 24.

9. Key risks and mitigation

Food safety: validate cyanogenic-compound reduction, microbiological controls and packaging before sale. Supply variability: use multi-site sourcing and seasonal planning. Commercial uncertainty: stage spending and conduct buyer trials before expansion. Cost escalation: obtain competitive quotations and maintain contingency. Institutional coordination: formalize roles and quarterly reviews. Environmental safeguards: prioritize responsible harvesting, wastewater management and appropriate disposal or use of processing residues.

10. Sustainability and next phase

The pilot will test unit economics, farm-gate incentives and viable business models. Expansion would depend on demonstrated safety, buyer demand, commercial margins, reliable raw-material supply and financing. The programme is designed to create reusable training, technical and market assets rather than a permanent subsidy-dependent operation.

11. Proposed immediate actions

Within 30 days of receiving an expression of interest, BIAJ proposes to convene a joint scoping meeting, confirm technical partners, validate the draft budget with quotations, identify candidate pilot sites and prepare a detailed implementation and procurement plan for approval.

12. Formal request

The Bamboo Industry Association of Jamaica respectfully requests the Ministry's consideration of the proposed J$39.67 million Government contribution, together with technical and institutional assistance, to establish the foundation for a safe, inclusive and commercially viable Jamaican bamboo shoots industry. BIAJ welcomes the opportunity to present the programme and refine its scope and budget jointly with the Ministry.

Submitted on behalf of the Board of Directors

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Chairman, Bamboo Industry Association of Jamaica

Date: _____

Appendix: The financial plan, workplan, partner commitments and regulatory pathway will be refined during project inception. No project funding or regulatory endorsement is represented as secured.

Annex A. Economic growth, trade and competitiveness case

This annex strengthens the economic justification for the proposed 24-month national bamboo shoots value-chain programme. Estimates below are illustrative and should not be represented as official GDP forecasts or verified sector accounts.

A1. Contribution to economic growth and development

The value chain would connect nurseries and farmers, harvesting and primary handling, transport, processing, packaging, quality assurance, distribution and retail. Domestic value added arises from the wages, operating surplus and production taxes generated at each stage, net of intermediate inputs. Potential development benefits include diversified farm incomes, rural employment, agro-processing investment, local technical capacity, entrepreneurship and opportunities for women and young people.

BIAJ should track hectares and growers supplying shoots; fresh-shoot tonnage; edible recovery and processing yield; domestic sales; exports; employment and wages; capital invested; and verified domestic value added. These metrics can demonstrate outcomes even while the industry remains too small to materially change national GDP.

A2. International trade benchmarks

Exporter	2022 preserved-shoot exports (US$)	Interpretation
China	$216 million	Large established exporter
Thailand	$18.3 million	Regional agro-processing exporter
Taiwan	$9.74 million	Specialized producer/exporter

Source: Data México, product profile HS 200591 (2022 global exporters). The figures measure trade in prepared/preserved bamboo shoots, not GDP or the full bamboo industry. FAO notes that fresh, chilled, brined and dried shoots can appear under other trade categories. Figures should not be scaled mechanically to Jamaica.

A3. Jamaica GDP contribution: production-based scenarios from BIAJ assessment

Using the BIAJ economic assessment (20 October 2026), the benchmark is Jamaica's 2025 nominal GDP of US$22,704,903,218, converted at an assumed J$160/US$ to J$3,632.8 billion. The estimates below are annual direct gross value added (GVA), not gross sales or official GDP forecasts. Comparing direct GVA with GDP is an indicative scale comparison, not a precise national-accounts estimate.

Metric	Conservative	Moderate / Ambitious
Productive hectares	200	600 / 1,500
Marketable shoots (tonnes/ha/year)	4	6 / 8
Annual fresh shoots (tonnes)	800	3,600 / 12,000
Direct GVA (J$/kg)	450	500 / 525
Annual direct GVA (J$ million)	360	1,800 / 6,300
Annual direct GVA (US$ million)	2.25	11.25 / 39.38
Share of 2025 GDP	0.010%	0.050% / 0.173%

These scenarios depend on unverified productive hectares, marketable yield and domestic value added per kilogram. They are planning assumptions rather than forecasts, and are not extrapolations from other countries. Actual sector GVA must be established through farm trials, verified processing costs, buyer demand and value-added accounts that avoid double counting.

Central scenario calculation: 600 productive hectares × 6 tonnes/hectare = 3,600 tonnes or 3.6 million kg annually. At J$500 direct domestic value added per kg, annual direct GVA is J$1.8 billion. Dividing by the J$3,632.8 billion GDP benchmark gives approximately 0.0495% (rounded to 0.050%).

Illustrative distribution of moderate-scenario J$1.8 billion GVA: cultivation and harvest 40% (J$720m); washing, trimming and preprocessing 15% (J$270m); thermal processing, canning and quality assurance 35% (J$630m); domestic logistics and distribution margins 10% (J$180m). These allocations are assumptions and must not be added to gross inter-stage sales.

Important qualification: The programme budget of J$49.588 million covers a 24-month validation and demonstration phase. The J$1.8 billion moderate scenario describes a hypothetical future annual industry at 600 productive hectares; it is not a return promised within the funded pilot period. Commercial scale-up requires additional private/public investment, confirmed demand and food-safety clearance.

Comparative interpretation: China, Thailand and Taiwan export figures in Annex A2 illustrate international trade activity, not their bamboo-shoot GDP shares. They cannot establish a reliable GDP percentage for Jamaica without comparable country-level value-added accounts. The Jamaican scenario is therefore independently constructed from production and unit-GVA assumptions.

Evidence needed to update the estimates: (i) commercial Bambusa vulgaris yields and seasonality; (ii) raw-to-edible recovery and canning yields; (iii) cyanogenic-compound, microbiological and can-seam tests; (iv) farm-gate and processing costs, imports and sales; (v) domestic/export demand; and (vi) two commercial seasons of sector accounting. Displacement of other crops and imported inputs may reduce net national gains.

A4. Trade, import substitution and competitiveness

- Import substitution: test the landed cost and consistent availability of imported shoots against the full cost of locally processed, safety-compliant products for Jamaican supermarkets, hotels and restaurants.
- Export diversification: test Caribbean and diaspora demand before committing to commercial-scale capacity; export potential depends on buyer specifications, shelf life, logistics and certification.
- Competitiveness: establish measurable benchmarks for raw-shoot recovery, cost per kilogram of edible product, processing yield, defect rate, energy and packaging cost, delivery reliability and market price.
- Product differentiation: examine Jamaican-origin branding and suitable products including canned shoots, while validating consumer acceptance and commercial demand.
- Market access: confirm tariff classifications and destination-market import, labeling and food-safety requirements; HS 200591 applies to certain prepared or preserved non-frozen shoots, not every bamboo shoot product.

A5. Food safety and technical gate

Commercialization must follow validated safe processing, including species and raw-material identification, reduction and testing of cyanogenic compounds, microbiological verification, container seam integrity and scheduled thermal processing appropriate to the product. The proposed SRC pilot using 24 A2 cans should be treated as a technical proof of concept, not evidence that a shelf-stable commercial product is approved. Independent testing and applicable Jamaican regulatory requirements must be confirmed before sale.

A6. Recommended economic workstream within the existing budget

Within the previously proposed programme budget, assign a documented economic-baseline and market-validation workstream: (1) Jamaican import and export data from STATIN and the Trade Board; (2) farmer supply and cost-of-production survey; (3) processor unit economics and sensitivity analysis; (4) local buyer interviews and export-market screening; (5) annual gross-output and value-added estimates. Reallocate existing research/market-testing provisions where feasible; any new cost should be separately approved rather than silently added to the funding request.

Sources and verification notes

- Data México, HS 200591 profile: https://www.economia.gob.mx/datamexico/en/profile/product/bamboo-shoots-prepared-or-preserved-otherwise-vinegar-or-acetic-acid-exc-frozen
- FAO, Bamboo trade statistics: https://www.fao.org/forestry/nwfp/statistics/bamboo/en

- World Bank GDP for Jamaica via FRED: https://fred.stlouisfed.org/series/MKTGDPJMA646NWDB
- FAO/INBAR discussion of wider bamboo sector (not shoots-only): https://www.fao.org/director-general/speeches/details/Signing-the-Memorandum-of-Understanding-between-FAO-and-the-International-Bamboo-and-Rattan-Organization-%28INBAR%29/en

Additional source for revised GDP scenario: BIAJ, “Jamaica’s Edible Bamboo Shoots Value Chain — Illustrative GDP Contribution and Economic Development Potential,” 20 October 2026 (user-supplied assessment). All scenario inputs are illustrative analyst assumptions.